In [1]:
```
pip install pandas scikit-learn
```

```
Requirement already satisfied: pandas in c:\users\system05\anaconda3\lib\site-packag
es (2.2.2)
Requirement already satisfied: scikit-learn in c:\users\system05\anaconda3\lib\site-
packages (1.5.1)
Requirement already satisfied: numpy>=1.26.0 in c:\users\system05\anaconda3\lib\site
-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\system05\anaconda3
\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\system05\anaconda3\lib\site-
packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\system05\anaconda3\lib\sit
e-packages (from pandas) (2023.3)
Requirement already satisfied: scipy>=1.6.0 in c:\users\system05\anaconda3\lib\site-
packages (from scikit-learn) (1.13.1)
Requirement already satisfied: joblib>=1.2.0 in c:\users\system05\anaconda3\lib\site
-packages (from scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\system05\anaconda3\l
ib\site-packages (from scikit-learn) (3.5.0)
Requirement already satisfied: six>=1.5 in c:\users\system05\anaconda3\lib\site-pack
ages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [3]:
```
data = pd.read_csv("C:/Users/System05/Downloads/heart.csv")
```

In [4]:
```
print(data.head())
```

```
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   52    1   0       125   212    0        1      168      0      1.0      2
1   53    1   0       140   203    1        0      155      1      3.1      0
2   70    1   0       145   174    0        1      125      1      2.6      0
3   61    1   0       148   203    0        1      161      0      0.0      2
4   62    0   0       138   294    1        1      106      0      1.9      1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0
```

In [8]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1025 non-null   int64
 1   sex       1025 non-null   int64
 2   cp        1025 non-null   int64
 3   trestbps  1025 non-null   int64
 4   chol      1025 non-null   int64
 5   fbs       1025 non-null   int64
 6   restecg   1025 non-null   int64
 7   thalach   1025 non-null   int64
 8   exang     1025 non-null   int64
 9   oldpeak   1025 non-null   float64
 10  slope     1025 non-null   int64
 11  ca        1025 non-null   int64
 12  thal      1025 non-null   int64
 13  target    1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

In [10]:
```
X = data.drop("target", axis=1)
y = data["target"]
```

In [11]:
```
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, reca
```

In [12]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_sta
```

In [15]:
```
print(len(X_train),len(X_test),len(y_train),len(y_test))
```

```
717 308 717 308
```

In [16]:
```
model = GaussianNB()
model.fit(X_train, y_train)
```

Out[16]:
```
▾ GaussianNB
GaussianNB()
```

In [17]:
```
y_pred = model.predict(X_test)
```

In [11]:
```
print("Predicted values:")
print(y_pred)
```

```
Predicted values:
[0 1 1 1 0 1 1 1 1 0 1 0 1 0 1 0 0 1 0 0 1 0 0 0 1 1 0 1 0 1 1 1 0 1 1 1 1
 0 0 1 1 0 1 1 0 1 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 0 1 1 0 0 1 1 1 0 0 1
 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 0 1 0 1 1 0 1 1 1 1 0 1 1 1 0 0 1 0 1 1 1 0
 1 0 1 1 1 0 1 1 1 0 0 1 0 1 0 1 1 1 1 0 1 0 0 1 0 1 0 1 1 0 0 0 1 0 1 0 0
 0 0 1 1 0 0 0 1 0 1 1 0 1 1 1 1 0 1 1 1 0 1 1 1 1 0 1 1 1 1 0 0 0 1 1 0 0
 0 1 1 1 0 0 1 0 0 0 0 0 0 0 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 0 1 1 1 1 0 0 1
 1 1 0 1 1 1 1 1 1 1 0 1 1 1 1 1 0 1 0 1 1 0 1 1 0 1 0 0 1 1 1 1 1 1 1 1 0
 0 0 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 1 1 1 1 0 0 1 0 0 1 1 0 1 0 0 0 1 0 0 1
 0 1 1 0 1 1 0 1 0 1 1 1]
```

In [12]:
```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7922077922077922
```

In [13]:
```
precision = precision_score(y_test, y_pred)
print("Precision:", precision)
```

```
Precision: 0.7344632768361582
```

In [14]:
```
cm = confusion_matrix(y_test, y_pred)
```

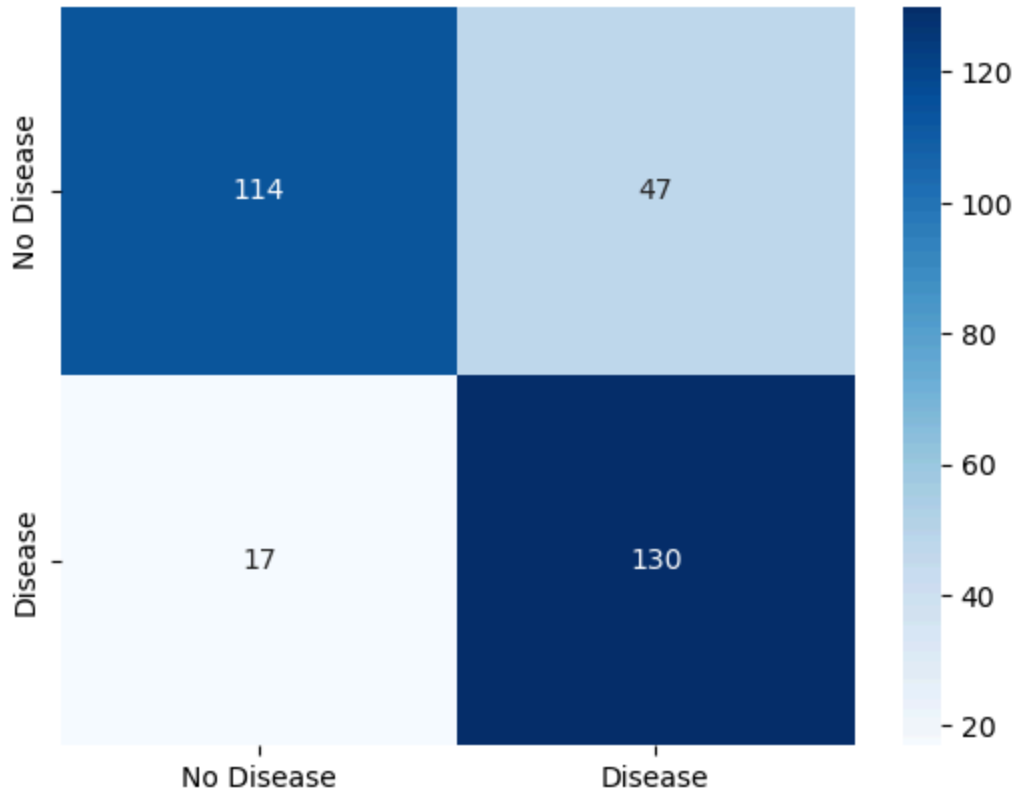
In [15]:
```
print("Confusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[114  47]
 [ 17 130]]
```

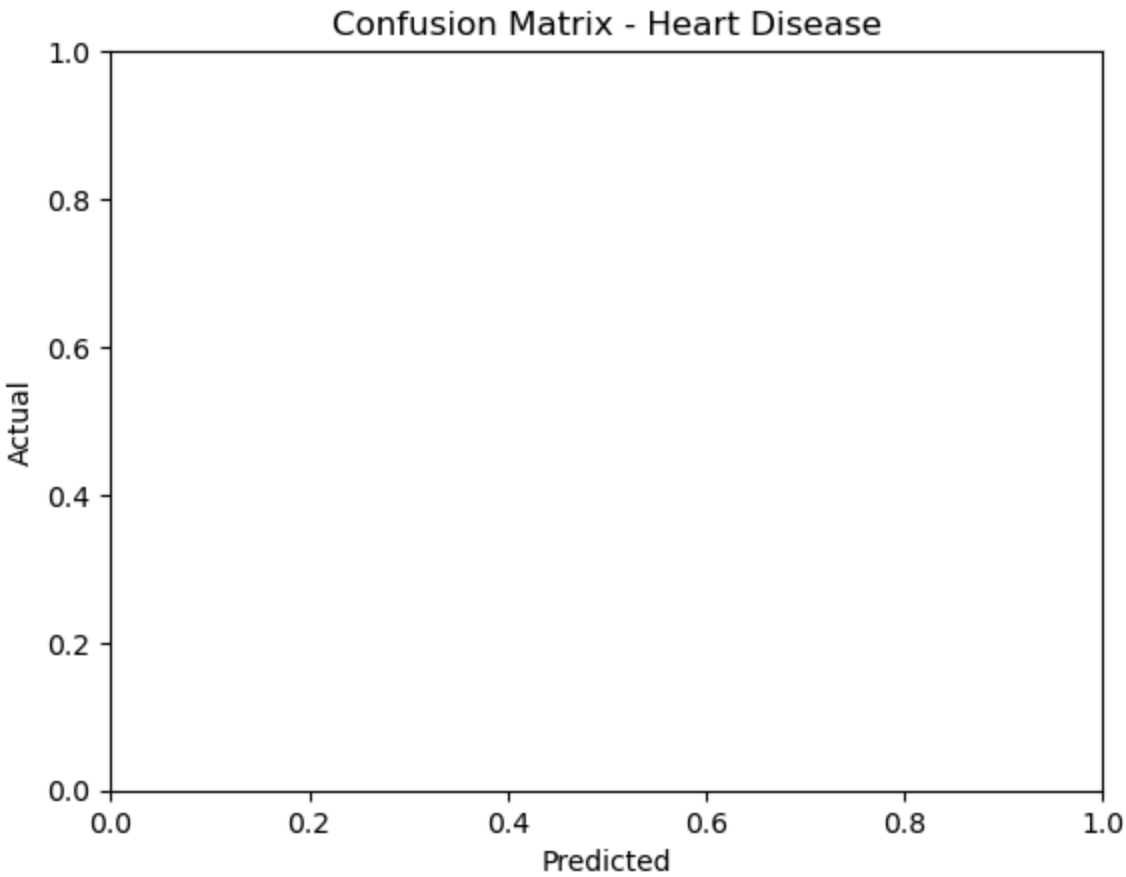
In [18]:
```
import seaborn as sns
```

In [19]:
```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=["No Disease", "Disease"],
            yticklabels=["No Disease", "Disease"])
```

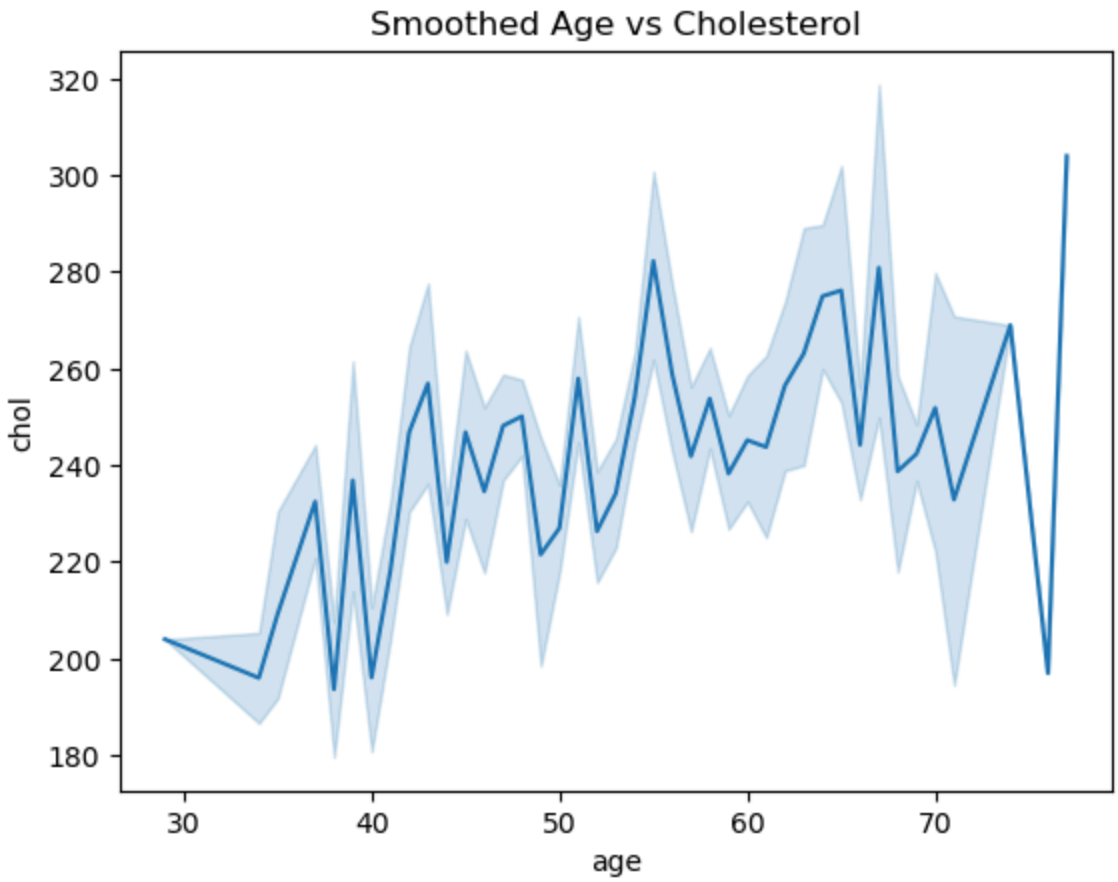
Out[19]:
```
<Axes: >
```



In [20]:
```
plt.xlabel("age")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Heart Disease")
plt.show()
```



In [23]:
```
sns.lineplot(x="age", y="chol", data=data)
plt.title("Smoothed Age vs Cholesterol")
plt.show()
```



In [25]:
```
sns.boxplot(x="target", y="chol", data=data)
plt.xlabel("Heart Disease")
plt.ylabel("Cholesterol")
plt.title("Cholesterol vs Heart Disease")
plt.show()
```

Cholesterol vs Heart Disease
500
400
300
200
Cholesterol
0
1
Heart Disease

In [ ]: